Experiment No. 7

Experiment Name

Customer Segmentation using K-Means Clustering

Aim

To implement the K-Means Clustering algorithm on a real-world dataset for customer segmentation and analyze customer groups based on purchasing behavior and demographic characteristics.

Purpose

Customer segmentation is a key application of unsupervised machine learning that helps organizations group customers with similar characteristics. By identifying distinct customer segments, businesses can design targeted marketing campaigns, improve customer engagement, optimize product recommendations, and enhance overall business performance.

In this experiment, students will use datasets such as the Mall Customer Segmentation Dataset, Online Retail Dataset, or Wholesale Customers Dataset to apply the K-Means clustering algorithm and interpret the resulting customer groups.

Prerequisites

- Basic knowledge of Python programming
- Familiarity with Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn
- Understanding of unsupervised machine learning concepts
- Knowledge of data preprocessing and feature scaling
- Jupyter Notebook or Google Colab

Steps

1. Select a suitable dataset such as the Mall Customer Segmentation Dataset, Online Retail Dataset, or Wholesale Customers Dataset.
2. Import the required Python libraries and load the dataset into a Pandas DataFrame.
3. Perform data preprocessing by handling missing values, removing duplicates, and selecting relevant features such as Age, Annual Income, and Spending Score.
4. Normalize or standardize the selected features to ensure that all variables contribute equally during clustering.

5. Determine the optimal number of clusters using the Elbow Method and validate the result using the Silhouette Score.
6. Apply the K-Means Clustering algorithm with the selected value of K to group customers into distinct clusters.
7. Visualize the clusters using scatter plots and highlight the cluster centroids for better interpretation.
8. Analyze the characteristics of each customer segment based on income, spending behavior, or purchasing patterns.
9. Compare the identified customer groups and suggest suitable marketing strategies for each segment.
10. Summarize the clustering results and discuss how customer segmentation can improve business decision-making.

Questions

1. What is clustering? How does it differ from classification?
2. Explain the working principle of the K-Means Clustering algorithm.
3. Why is feature scaling important before applying K-Means clustering?
4. What is the Elbow Method, and how does it help determine the optimal number of clusters?
5. What is the Silhouette Score? How is it used to evaluate clustering performance?
6. Why is K-Means considered an unsupervised machine learning algorithm?
7. Mention any four real-world applications of customer segmentation.
8. What are the limitations of the K-Means algorithm?
9. How can businesses use customer segmentation to improve marketing and customer retention?
10. Compare K-Means Clustering with Hierarchical Clustering based on their working principles and applications.